

# **Data Analysis Strategy for Green Jobs & Sustainability Projects**

*Week 1 Deliverable — Strategic Plan Document*

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# 1. Executive Summary

This document presents the strategic plan that will guide all data-analysis activity for the Green Jobs & Sustainability internship track. Green jobs and sustainability metrics sit at the intersection of workforce data, energy statistics, and environmental indicators, and no single public dataset covers all three well. The strategy therefore takes a triangulation approach: combine labour-market sources (ILOSTAT, national statistics), environmental and energy sources (IRENA, IEA, World Bank), and India-specific sources (data.gov.in, NITI Aayog, CEEW) to build a defensible, repeatable picture of how the green economy is growing and where the gaps are.

The plan defines five research questions, a set of seven core KPIs spanning workforce, economic, and environmental dimensions, and a five-week execution roadmap that carries this strategy through data collection, exploratory analysis, predictive modeling, and a final synthesis report. Because no proprietary databases are available, every source identified below is free and publicly accessible, and the plan explicitly documents how data-quality gaps (definitional inconsistency, reporting lag, geographic granularity) will be handled rather than ignored.

## 2. Introduction & Background

The International Labour Organization defines a green job as decent work that contributes to preserving or restoring the environment, spanning traditional sectors (agriculture, manufacturing, construction) as well as emerging ones (renewable energy, e-mobility, waste management, sustainable finance). Global policy momentum — the Paris Agreement, the EU Green Deal, the U.S. Inflation Reduction Act, and India's National Green Hydrogen Mission and Panchamrit climate commitments — is accelerating investment into this segment, which in turn is reshaping labour demand and skill requirements.

In India specifically, the transition is visible in rising installed renewable capacity, state-level skilling missions run through the Skill Council for Green Jobs, and growing corporate ESG (Environmental, Social, Governance) reporting obligations. Each of these generates data — but scattered across government portals, multilateral agencies, and research institutes rather than one consolidated source. This creates a genuine analytical opportunity: a well-designed strategy that pulls together the right combination of public sources can produce insights (which sectors are growing fastest, where skill gaps are largest, how renewable capacity growth correlates with job creation) that are otherwise hard to see in any single dataset.

## 3. Objectives & Research Questions

The primary objective of this internship track is to build a repeatable analytical framework — not a one-off report — that can quantify green-job growth, benchmark sustainability performance, and forecast near-term trends using only publicly available data. The framework is organized around five research questions:

- Which sectors and Indian states are generating the largest number of green jobs, and how fast are they growing?
- How does growth in renewable energy capacity correlate with green employment growth over the same period?
- Where are the largest gaps between green-job demand (postings/roles) and available green-skilled talent?
- Which sustainability KPIs (emissions intensity, SDG index score, renewable share) most strongly track with employment outcomes?

- Based on current trends, what does the green-jobs and sustainability trajectory look like over the next 2–3 years?

## 4. Analytical Framework

### 4.1 Key Metrics (KPIs)

Seven core metrics anchor the analysis across the workforce, economic, and environmental dimensions:

| Metric                               | Definition                                                                          | Category                 |
|--------------------------------------|-------------------------------------------------------------------------------------|--------------------------|
| Green job count & growth rate        | Number of jobs classified as green (ILO definition) and year-over-year change       | Workforce                |
| Installed renewable capacity (MW)    | Cumulative solar, wind, and other renewable capacity by region                      | Environmental / Economic |
| Green skill-gap index                | Ratio of green job postings to candidates with matching skills                      | Workforce                |
| Investment in green sector (USD/INR) | Public and private capital directed to renewable energy and sustainability projects | Economic                 |
| Carbon emission intensity            | Emissions per unit of GDP or per capita, tracked over time                          | Environmental            |
| % of workforce in green occupations  | Share of total employment classified as green, by sector/state                      | Workforce                |
| SDG India Index score                | State-wise composite sustainability performance score                               | Environmental / Policy   |

### 4.2 Analytical Methods

The framework applies methods in increasing order of complexity, matched to the internship's weekly cadence:

- Descriptive statistics — central tendency and dispersion for each KPI, by year, sector, and state.
- Trend / time-series analysis — year-over-year growth rates in green employment and renewable capacity.
- Correlation analysis — relationship between renewable capacity growth and job creation; between investment and emissions intensity.
- Regression / forecasting models — simple linear and time-series models (e.g., ARIMA or linear trend extrapolation) to project 2–3 year outlooks.
- Geographic comparison — state-wise benchmarking using the NITI Aayog SDG India Index and MNRE capacity data.

### 4.3 Tools

- Python (pandas, NumPy, matplotlib/seaborn, scikit-learn) for cleaning, EDA, and modeling.
- Power BI Desktop for interactive dashboards and stakeholder-facing visuals.
- Google Sheets for lightweight tracking and quick pivot summaries where Power BI is not needed.

## 5. Data Collection Strategy — Public Data Sources

Sources were selected against four criteria: credibility (government, multilateral, or peer-reviewed origin), currency (updated at least annually), licensing (open for public/academic use), and relevance to at least one of the seven core KPIs. The table below is the working source list for Weeks 2–4; it will be narrowed to 2–3 primary sources per task to keep collection effort proportionate to the available time.

| Source                                                 | Type of Data                                                                | Coverage & Update Cycle                  | Access                               |
|--------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------|--------------------------------------|
| ILOSTAT<br>(International Labour Organization)         | Employment by sector/occupation, green job estimates, labour force surveys  | Global, country-level; updated annually  | Free — web portal & bulk CSV/API     |
| World Bank Open Data & Climate Change Knowledge Portal | GDP, energy use, emissions, climate risk indicators                         | Global, 200+ countries; updated annually | Free — CSV/API                       |
| UN SDG Indicators Database                             | Indicators for SDG 7 (energy), 8 (work), 12 (consumption), 13 (climate)     | Global & national; updated annually      | Free — data portal                   |
| IRENA<br>(International Renewable Energy Agency)       | Installed renewable capacity, renewable-sector employment (Annual Review)   | Global & country-level; updated annually | Free — reports & statistics queries  |
| IEA (International Energy Agency)                      | Energy production, consumption, and transition indicators                   | Global; updated annually                 | Free tier + paid detailed datasets   |
| data.gov.in (Govt. of India Open Data)                 | MNRE renewable capacity, MoEFCC environmental data, state energy statistics | India, state-level; varies by dataset    | Free — open government data          |
| NITI Aayog SDG India Index                             | State/UT-wise sustainability & goal-wise performance scores                 | India, state-level; updated periodically | Free — published reports & dashboard |
| CEEW (Council on Energy, Environment and Water)        | India-focused energy transition and green jobs research                     | India; updated with each study           | Free — public reports                |
| Our World in Data                                      | Curated, cleaned datasets on energy, emissions, and environment             | Global; continuously updated             | Free — CSV download                  |
| Kaggle / Google Dataset Search                         | Community and curated datasets tagged sustainability, energy, ESG           | Varies by dataset                        | Free — direct download               |

Where a metric cannot be sourced directly (for example, granular district-level green employment in India), the strategy is to construct a clearly labeled proxy or simulated estimate — built from a documented, defensible assumption (e.g., allocating state-level MNRE capacity data proportionally using workforce census ratios) — rather than to leave the KPI blank. Every proxy will be flagged as such in the final report so stakeholders can distinguish observed data from modeled estimates.

## 6. Data Analysis & Visualization Approach

Analysis and visualization work is sequenced to match the internship's weekly structure, so each week's output feeds directly into the next rather than being a standalone exercise:

| Week   | Phase                             | Primary Focus                                                                                      |
|--------|-----------------------------------|----------------------------------------------------------------------------------------------------|
| Week 1 | Strategy                          | Define scope, research questions, KPIs, and this strategic plan (current deliverable).             |
| Week 2 | Data Collection & Prep            | Identify sources, extract data, handle missing values, standardize formats.                        |
| Week 3 | Exploratory Analysis              | Descriptive statistics, correlation checks, trend and distribution analysis, first visualizations. |
| Week 4 | Statistical & Predictive Modeling | Regression/time-series modeling to forecast green job and sustainability trends.                   |
| Week 5 | Synthesis & Final Report          | Consolidate findings into a strategic report with recommendations for stakeholders.                |

Planned visualization types include:

- Line charts for year-over-year green-job and renewable-capacity growth trends.
- Bar charts comparing sectors or states on a single KPI (e.g., installed capacity by state).
- Scatter plots and correlation heatmaps to test relationships between KPIs (e.g., capacity growth vs. job growth).
- Choropleth-style state maps (Power BI) for the SDG India Index and green-employment share.
- A consolidated Power BI dashboard in Week 4–5 combining the above into one stakeholder-facing view.

## 7. Anticipated Challenges & Mitigation

- Inconsistent “green job” definitions across sources — adopt the ILO definition as the baseline and explicitly document where a source's classification diverges from it.
- Reporting lag (most annual datasets trail by 12–18 months) — supplement historical data with the most recent industry and government reports for directional, qualitative context.
- Geographic granularity gaps, especially at the district level in India — use documented proxy estimates as described in Section 5, always flagged as modeled rather than observed.
- No access to proprietary or paid databases — restrict the source list to free, public datasets and treat this as a design constraint rather than a limitation to work around.

- Time budget (30–35 hours/week across research, analysis, and documentation) — prioritize 2–3 primary sources per weekly task instead of attempting exhaustive coverage.

## **8. Conclusion**

This strategy gives the internship track a clear, defensible path from research question to final report: five questions, seven KPIs, ten vetted public sources, and a five-week execution plan that builds analytical complexity week over week. It is designed to stand on its own as a blueprint — any analyst following this document should be able to replicate the approach on a similar green-jobs or sustainability dataset. The next deliverable (Week 2) will apply this strategy to identify and prepare the specific datasets needed for exploratory analysis.